PNAS RESEARCH ARTICLE | DEVELOPMENTAL BIOLOGY

Cellular heterogeneity and dynamics of the human uterus in healthy premenopausal women

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Edited by Thomas Spencer, University of Missouri, Columbia, MO; received March 8, 2024; accepted September 19, 2024

The human uterus is a complex and dynamic organ whose lining grows, remodels, and regenerates every menstrual cycle or upon tissue damage. Here, we applied single-cell RNA sequencing to profile more the 50,000 uterine cells from both the endometrium and myometrium of five healthy premenopausal individuals, and jointly analyzed the data with a previously published dataset from 15 subjects. The resulting normal uterus cell atlas contains more than 167K cells, representing the lymphatic endothelium, blood endothelium, stromal, ciliated epithelium, unciliated epithelium, and immune cell populations. Focused analyses within each major cell type and comparisons with subtype labels from prior studies allowed us to document supporting evidence, resolve naming conflicts, and propose a consensus annotation system of 39 subtypes. We release their gene expression centroids, differentially expressed genes, and messenger Ribonucleic Acid (mRNA) patterns of literature-based markers as a shared community resource. We identify multiple potential progenitor cells: compartment-wide progenitors for each major cell type and potential cross-lineage multipotent stromal progenitors that may replenish the epithelial, stromal, and endothelial compartments. Furthermore, many cell types and subtypes exhibit shifts in cell number and transcriptomes across different phases of the menstrual cycle. Finally, comparisons between premenopausal, postpartum, and postmenopausal samples revealed substantial alterations in tissue composition, particularly in the proportions of stromal, endothelial, and immune cells. The cell taxonomy and molecular markers we report here are expected to inform studies of both basic biology of uterine function and its disorders.

uterine stem cells | endometrium | myometrium | uterine homeostasis | menstrual cycle

The menstrual cycle is an essential feature of reproductive physiology in many mammals, including humans, higher-order primates, and certain rodents and bats. It begins with the shedding of the endometrial lining and ends with its complete repair, making the uterus one of the most dynamic organs (1). This cyclical regeneration process occurs up to 400 times during a woman's reproductive lifespan (2–4). Approximately every 28 d, the cycle initiates in the proliferative phase, marked by a significant thickening of the endometrium. Following ovulation, the endometrial lining undergoes decidualization in response to progesterone secreted by the corpus luteum in the ovary. This process prepares the endometrium for the potential implantation of an embryo. In the absence of embryo implantation, the lining sheds back to the basalis layer; and the uterus undergoes rapid involution, setting the stage for the next cycle (5). This continuous, dynamic remodeling of the uterus is linked to systemic hormonal regulation and the ovulation cycle. The cyclic changes in the endometrium and myometrium, along with their regenerative capacities, raise important questions about the cellular dynamics during these phases and the identity of the stem or progenitor cells responsible for maintaining tissue homeostasis. After decades of research, several stem/progenitor cells have been identified that are proposed to maintain the stromal and epithelial compartments (4–11), respectively, suggesting that the myometrium and endometrium are sustained by compartment-specific progenitors.

To build on these foundational studies, single cell RNA sequencing (scRNA-seq) of human endometrium samples has been used to delve deeper into uterine tissue heterogeneity and homeostasis (11–13), with each reporting a cell annotation system that covered diverse stromal, epithelial, endothelial, and immune cell types. Although each dataset identifies likely homologous populations, an integrated analysis is needed to synthesize the multiple taxonomies. Furthermore, most of the analyzed samples were from endometrial biopsies, thus the knowledge of the basalis endometrium and myometrium is still lacking. To address these gaps, we performed scRNA-seq analyses on full-thickness uterus samples from five premenopausal women, followed by integration with publicly available data for 15 samples from Garcia-Alonso et al. (12) and Wang et al. (11). This effort led to a consensus cell atlas that includes cell populations enriched in specific compartments. In

Significance

Our integrated analyses of 20 single-cell RNA sequencing (scRNA-seq) human uterus datasets reveals 39 cell subtypes across four major cell types, laying the groundwork for establishing a stable, consensus cell atlas of the human uterus. Furthermore, we identify multiple computationally predicted progenitor populations for each of the major cell compartments, as well as potential cross-compartment, multipotent progenitors. While the function and interactions of these cell populations remain to be validated through future experiments, the markers and their "dual characteristics" that we describe will serve as a rich resource to the scientific community. Finally, a comparative analysis across menstrual cycle phases reveals many notable shifts in cell populations and changes in cell type-specific transcriptomes, recapitulating old findings and revealing biological insights.

The authors declare no competing interest.

This article is a PNAS Direct Submission.

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This article contains supporting information online at https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2404775121/-/DCSupplemental.

Published October 29, 2024.

all, we identified five major cell types and 39 subtypes: with many phase-regulated genes and pathways across compartments and cell types. Computational analyses of mRNA "velocity" and "pseudotime" revealed putative compartment-specific progenitors as well as dual-lineage progenitors, such as a myometrial multipotent mesenchymal progenitor capable of contributing to the epithelium and endothelium.

Results

ScRNA-seq of Five Full-Thickness Uterine Samples Reveals Five Major Cell Types and Their Functional Subtypes. We performed scRNA-seq on full-thickness uterus samples from five healthy donors: three in the Proliferative phase, and one each in the Mid-Secretory and Late Secretory phase, respectively (Fig. 1*A* and Dataset S1*A*). After separating the endometrium and myometrium layers, we analyzed >50,000 cells that passed quality control filtering (*Materials and Methods* and *SI Appendix*, Fig. S1*A*). Clustering analyses for each of the endometrial and myometrial samples revealed 9 to 11 clusters, with profiles highly correlated across samples (*SI Appendix*, Fig. S1*B*). This observation allowed us to combine samples (*Materials and Methods*), resulting in an integrated dataset containing five major cell clusters (Fig. 1*B*). By using both statistically generated (Fig. 1*C* and Dataset S3) and established literature-based markers (Fig. 1*D*) we annotated the five major clusters as lymphatic endothelial (PROX1, NRP2, PDPN, FXYD6, CD36, CAVIN2), blood endothelial (VWF, PECAM1), stromal (MYH11, ACTA2, COL1A1, THY1, PDGFRB, DES), epithelial (consisting of both ciliated epithelial—Epithelial Cell Adhesion Molecule (EPCAM), CAPS, FOXJ1, and nonciliated epithelial cells—EPCAM, PAX8), and immune (KIT, PTPRC, CD3, RUNX3, S100A9, FOLR2, CD86) cells. As expected, myometrium samples contributed more of the blood endothelial and stromal cells, while the endometrium samples were dominated by epithelial cells (*SI Appendix*, Fig. S1*C* and Dataset S1*B*). Lymphatic endothelial cells (LECs) and immune cells showed variable contributions from different samples (*SI Appendix*, Fig. S1*C* and Dataset S1*B*).

We subsequently performed iterative reclustering on each major cell type to identify finer subtypes, except for LECs due to the small number of cells recovered. We identified four blood endothelial, eight stromal, nine epithelial (one ciliated and eight nonciliated), and six immune cell subtypes (Fig. 1*E*), for 27 subtypes (28 if counting lymphatic endothelial). Their cluster centroids and differentially expressed (DE) genes, for both the major cell types and the subtypes, were included in Datasets S2 and S3.

Joint Analyses of 20 Samples from Three Studies Led to a Consensus Cell Atlas of the Human Uterus. While a natural next step is to biologically annotate the 27 subtypes, we recognized that several prior studies had attempted to define cell types in the human uterus, with independent and sometimes discordant results (11–13). To reduce confusion among different versions of uterus cell annotation, and to systematically develop a strategy to build a stable, consensus cell atlas, we decided to focus on a multistudy combined clustering and reclustering analysis. Since the subjects in Tan et al. (13) were either on hormone treatment or diagnosed with endometriosis, we only used their cluster centroids for comparison. For the 15-donor data by Wang et al. (11) and Garcia-Alonso et al. (12) we accessed the individual cell-level data as well as the cell type annotations reported by Garcia-Alonso et al. After merging data from the 20 donors (Dataset S4*A*), containing a total of 167,910 cells, and batch-correcting by donor, we arrived at five major cell types that were contributed by all 20 samples (Fig. 2*A* and *SI Appendix*, Fig. S2*B*). Focused reclustering identified eight blood endothelial, 11 stromal, nine epithelial (one ciliated and eight nonciliated), and 10 immune cell subtypes (Fig. 2*B*), for a total of 38 subtypes (39 if counting lymphatic endothelial). The centroids of major cell types and subtypes were included in Dataset S5, with DE genes shown in Dataset S6. Fig. 2*C* shows the literature-based marker genes (with underlying data in Dataset S7); and Fig. 2*D* shows the relative cell proportion over the subtypes, for the four cell compartments separately and across menstrual phases.

To develop the consensus nomenclature for the stromal, epithelial, blood endothelial, and immune cells, we applied three types of metrics to examine the level of concordance between the past and new annotation results. First, since the cells from the 15 donors have been clustered by Garcia-Alonso et al. and assigned names, and they are clustered again in this study in a 20-donor joint analysis, we compare cell cluster assignments using *cross-tabulation of cell counts* to indicate the correspondence and stability between two annotation results. Second, we calculate *pairwise centroid–centroid correlation matrices* to measure the correspondence among subtypes. Third, we produce panels of DE genes from one annotation and plot *gene-by-centroid heatmaps* for centroids from another annotation (and vice versa), to see if the DE genes remain specific in the latter (Note: comparisons with Tan et al. only used second and third approaches). By examining these complementary lines of evidence, we were able to propose standardized nomenclature and, importantly, document the uncertainty in some results (*SI Appendix*, *Summaries and Discussions* and Figs. S3–S6).

Identification of 11 Stromal Cell Subtypes. Focused reclustering of the ~89K stromal cells identified 11 subclusters, provisionally labeled as Stro-1 to Stro-11 (Fig. 2*B* and *SI Appendix*, *Summaries and Discussions* and Fig. S3). The expression values for literature-based markers and DE genes are in Datasets S7 and S6, respectively. Briefly, the 11-stromal subtypes correspond to:

Stro-1, Stro-2, and Stro-3 are three distinct types of perivascular (PV) cells. While all three express PV markers (MYH11, PDGFRB, MCAM, RGS5, SUSD2; Fig. 2*C* and Datasets S6 and S7), each has its unique distinguishing markers. For example, Stro-1 expresses higher levels of FOXC1, DEPP1, and CDKN1A; Stro-2 expresses higher levels of CTGF, HOPX, PRKCDBP (CAVIN3), and ATP5L; and Stro-3 has the highest expression of STEAP4 (Fig. 2*C* and Datasets S6 and S7). Gene ontology analysis suggests that Stro-1 cells are involved in secreting extracellular matrix (ECM), and likely impacting the rigidity or mobility of endothelial cells (14, 15), while Stro-2 is actively involved in angiogenesis. Stro-3 cells appear to have an immunoregulatory role by modulating interferon signaling, consistent with the previously described anti-inflammatory and antiapoptotic role of STEAP4 endothelial cells in retinal vasculature and other organs (16). Further, our ligand–receptor analysis suggests that PV STEAP4 cells indeed communicate with macrophages (Imm-8), NK cells (Imm-9), and CD8 cells (Imm-3) (see *SI Appendix*, Fig. S7*D* below).

Stro-4 is named tissue-resident vascular progenitor. These cells express some endothelial markers such as VWF and PECAM1 (Fig. 2*C* and Datasets S6 and S7) but lack HEG1 and FLI1. Immunostaining confirmed the presence of VWF^+, $PECAM1^+$, but $FLI1^-$ stromal cells (*SI Appendix*, Fig. S3*G*). Additionally, they express CD141 (THBD), CD144 (CDH5), CD105 (ENG), MCAM (Fig. 2*C* and Datasets S6 and S7), markers for tissue-resident vascular progenitor cells, which are known to reside in the intima of the vessels in individual organs (17). As we show below (Fig. 3*A*), this population appears to bridge with endothelial cells, especially Endothelial-to-Mesenchymal Transition (see below). Therefore, we propose that Stro-4 is a tissue-resident vascular progenitor, suggesting that the endothelial cells within the uterus have two sources: a bone marrow origin and a stromal-cell origin. These findings support a recent model that questions if all endothelial cells come from bone marrow (17, 18).

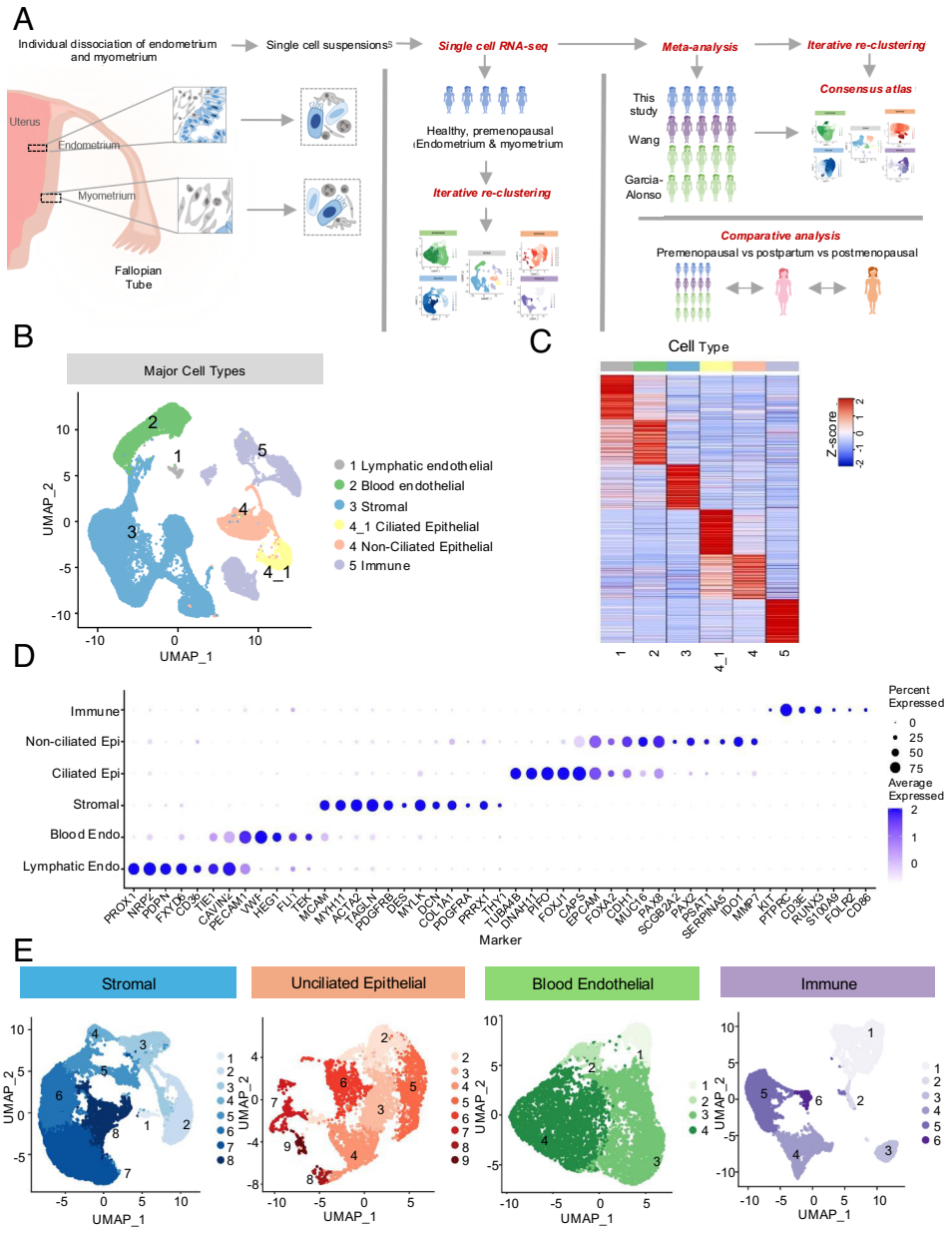


Fig. 1. Cell types and markers identified from scRNA-seq analysis of five donors. (*A*) Study overview, including sample and data collection, integration with prior data from 15 donors to generate a consensus atlas, and comparison with a postpartum and a postmenopausal sample. (*B*) Identification of five major cell types from global clustering, visualized in Uniform Manifold Approximation and Projection (UMAP). The epithelial cells are split into ciliated (4_1) and unciliated epithelial cells (4). (*C*) Top 150 differentially expressed genes for each major cell type, shown as a gene-centroid heatmap for per-gene standardized values (data in Dataset S3*A*). (*D*) Dot plot of literature-based marker genes used to annotate the major cell types. (*E*) Identification of cell subtypes by focused reclustering of each major cell type, shown in UMAP, from *Left* to *Right*, for stromal, unciliated epithelial, blood endothelial, and immune cells.

Stro-5, RUNX3+ fibroblast, expresses immune markers RUNX3 and CD3E, and has similarities with Imm-2: Fibrocytes (see Fig. 3*A* below). To verify its stromal-immune hybrid character, we performed gene–gene correlations in individual Stro-5 cells and show that these cells coexpress both stromal markers (THY1, PDGFRA, COL1A1) and immune markers (RUNX3, PTPRC, GNLY, CD3E) (*SI Appendix*, Fig. S3*H*). Immunostaining confirms the presence of RUNX3$^+$, ACTA2$^+$, and PDGFRA$^+$ triple-positive cells (*SI Appendix*, Fig. S3*G*). RUNX3 functions as a tumor suppressor gene in the endometrium and other tissues and is reduced in endometrial cancer (19, 20). Consistent with a possible role in immune function, Stro-5 expresses the highest level of HLA-A and C (Fig. 2*C* and Datasets S6 and S7), which are class-I histocompatibility antigens that present antigen peptides to cytotoxic T cells; we therefore postulate that Stro-5 may be a dedicated antigen-presenting fibroblast population.

Stro-6, uterine smooth muscle cells (uSMC), specifically expresses SMC markers ACTA2 and DES and has moderate expression of MYH11; note that MYH11 is highest in Stro-1/2 (Fig. 2*C* and Datasets S6 and S7). Unlike the PV cells (Stro-1/2), Stro-6 has much lower expression of MCAM and PDGFRB.

Stro-7, C7 Fibroblast, is more abundant in myometrium than endometrium samples (Fig. 2*D*), possibly due to the myometrium samples partially retaining the basal layer of the endometrium. Notably, C7, the specific marker for this population, was highly expressed in the basal layer of the endometrium as shown by a spatial transcriptomic analysis (12). Receptor–ligand analysis suggests that C7 fibroblasts communicate broadly, both within stroma and across other compartments (*SI Appendix*, Fig. S7*D*).

Stro-8, Stro-9, and Stro-10 are labeled as decidualized fibroblast cells 1 to 3, and they account for 25% of all stromal cells (40K/167K). Within this fibroblast pool, Stro-8 accounts for 8.6%, Stro-9 for 53.24%, and Stro-10 for 38.6%. These cells express varying levels of downstream targets of the cyclic Adenosine Monophosphate (cAMP) and insulin signaling, such as FOXO1, WNT4, PRL, IRS2, IGFBP1, and IGFBP3 (21–25) (Fig. 2*C* and Datasets S6 and S7). FOXO1 is a core decidual transcription factor (TF) that controls cell cycle exit of endometrial fibroblast cells and activates decidual marker genes, such as WNT4, PRL, and IGFBP1 (22, 23, 26). Following cell cycle exit, decidualizing endometrial fibroblast cells first mount a transient pro-inflammatory response, known as the senescence-associated secretory phenotype (SASP)

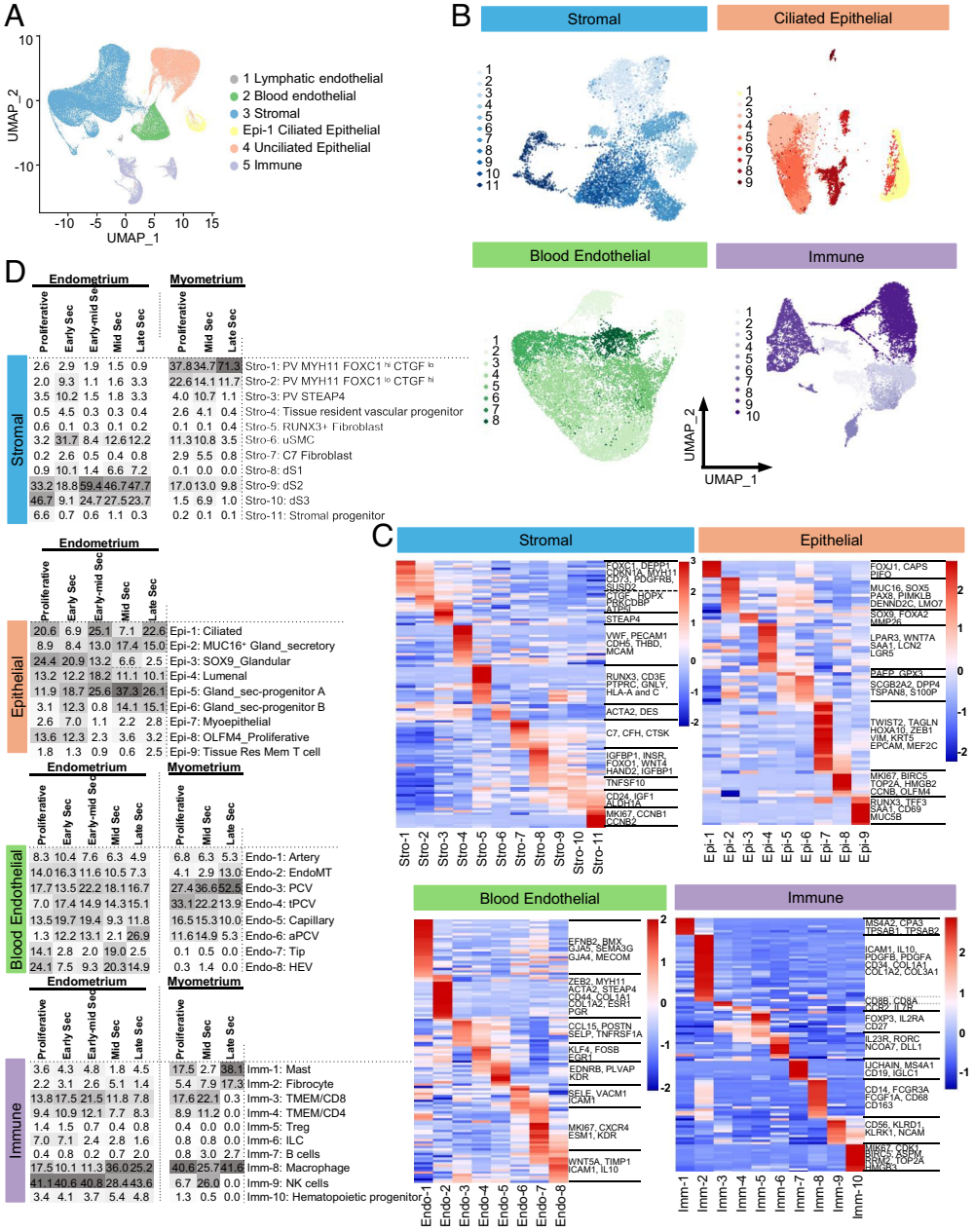


Fig. 2. Construction of a 20 sample consensus cell atlas. (*A*) Identification of five major cell types, shown in UMAP. Epithelial cells are split into the ciliated (Epi-1) and unciliated cells. (*B*) Identification of cell subtypes by focused reclustering of the four major cell type, shown in separate UMAP plots. The panel for stromal cells used a randomly down-sampled set of stromal cells for better clarity, while the corresponding UMAP for all stromal cells is shown in *SI Appendix*, Fig. S7*B*. (*C*) Subtype-specific expression of literature-based marker genes used to annotate the cell subtypes. Shown are per-gene standardized values from the 20-sample integrated analysis (data in Dataset S7). (*D*) Comparison of the relative abundance of cell subtypes across samples of different cycle phases. Subtype fractions are calculated for the four major cell types separately (*Top* to *Bottom*) and averaged over the samples of the same tissue source (endometrium or myometrium) and the same cycle phase (Dataset S4). Values were colored with shades of gray.

(27), characterized by the transient secretion of pro-inflammatory cytokines such as CXCL1&8, CXCL14, and IL6 (26, 28). Here, we show that these pro-inflammatory cytokines are most highly expressed in Stro-8, but decrease in Stro-9 and Stro-10. We therefore consider Stro-8 to be a SASP+ decidual senescent fibroblast, while Stro-9 and Stro-10 are transitioning and terminally decidualized cells, respectively.

Stro-11, stromal progenitor, is notable for its uniquely high expressions of cell cycle regulator MKI67 and the cyclins CCNB1-2 (Fig. 2*C* and Datasets S6 and S7) and is more prevalent in proliferative-stage endometrium samples (Fig. 2*D*). In addition to being the most actively cycling stromal population, RNA velocity and pseudotime analysis identified Stro-11 as a multipotent progenitor upstream of other stromal cell subtypes (see below Fig. 3*B* and Datasets S6 and S7).

To validate the presence of the 11 stromal subtypes in the human uterus, we performed flow cytometry analysis of a full thickness uterus sample using a combination of seven antibodies [Melanoma Cell Adhesion Molecule (MCAM), CD34, CD24, CD90, CD31, HOPX, CDKN1A]. The seven protein markers revealed 11 groups of cells (*SI Appendix*, Fig. S3*I*), with patterns comparable to our 11 stromal subtypes defined by scRNA-seq.

In sum, we identified three PV cells (Stro-1,2,3), which are different from the more abundant uSMCs (Stro-6). Additionally, we identified an unexpected tissue-resident vascular smooth muscle progenitor (Stro-4), which has similarities to endothelial cells (see Fig. 3*A* below). Stro-5 (RUNX3+ fibroblast) has both stromal and immune cell characteristics; and Stro-7 is a C7 fibroblast that, like other fibroblasts, may recruit various immune cell types (29). Finally, Stro 8-10 correspond to three distinct decidualized stromal cells; and Stro-11 is a potential proliferative multipotent endometrial stromal cell progenitor (further described in Fig. 3 *A* and *B* and Datasets S6 and S7).

Identification of Nine Epithelial Cell Subtypes. The epithelium layer in the endometrium consists of two major epithelial cell subtypes: glandular and luminal (30). To examine epithelial cell heterogeneity, we reclustered the ~47K cells from the 20 donors

and identified nine subclusters: Epi-1 to Epi-9 (Fig. 2*C*). The stability of these clusters, and their annotation, are detailed in *SI Appendix*, Fig. S4 and *Summaries and Discussions*.

Epi-1 is Ciliated population as it expresses classic ciliated markers i.e. FOXJ1, CAPS, and PIFO (Fig. 2*C* and Dataset S7). Epi-2, Glandular Secretory, expresses MUC16, SOX5, PAX8, RIMKLB, DENND2C, and LMO7 (Fig. 2*C* and Datasets S6 and S7); many of which are previously described glandular secretory cell markers (31). Epi-3, Sox9 Glandular, specifically expresses FOXA2 and MMP26, two known glandular markers (7, 32), as well as SOX9.

Epi-4 is named Luminal, as it expresses LPAR3, LGR5, WNT7A, SAA1, and LCN2 (Fig. 2*C* and Datasets S6 and S7), which are implicated in luminal epithelial cell function (33–35). For example, LPAR3 is necessary for successful embryo implantation and uterine receptivity in mice (36), and WNT7A is secreted by luminal epithelial cells to support postmenstrual endometrial regeneration (35, 37).

Epi-5 and Epi-6 are Glandular secretory-progenitor A and B, respectively. They both highly express PAEP, GPX3, and SCGB2A2, known glandular secretory markers; yet Epi-5 expresses higher levels of S100P, while Epi-6 expresses higher levels of TSPAN8, DPP4, as well as a moderate level of ciliate cell markers: FOXJ1, PIFO, and CAPS (Fig. 2*C* and Datasets S6 and S7). Pseudotime and velocity analysis suggests that Epi-5 is an upstream progenitor for Epi-2 & Epi-3, while Epi-6 is likely an upstream progenitor for Epi-1 (Fig. 3*C*).

Epi-7 specifically expresses EPCAM along with WT1, TWIST2, TAGLN, SNAI2, ZEB1, VIM, suggesting that this is a Myoepithelial cell population (Fig. 2*C* and Datasets S6 and S7). We confirmed the presence of these flattened epithelial cells by costaining EPCAM and WT1 (*SI Appendix*, Fig. S4*H*). Epi-7 is the only epithelial cell population that expresses AR, while ESR1 is broadly expressed in all epithelial subtypes, and PGR is specifically high in Epi-3 and Epi-7,8,9 (*SI Appendix*, Fig. S2*C*). Interestingly, Epi-7 appears to resemble a population we previously identified in the fallopian tube (38) (*SI Appendix*, Fig. S4*G*).

Epi-8, OLFM4_proliferative, expresses specifically high levels of TF HMGB2, OLFM4, and cell cycle genes TOP2A, MKI67, BIRC5, and CCNB. While it matches a cell type named "Sox9_prolif" in Garcia-Alonso et al. (*SI Appendix*, Fig. S4 *B*, *C*, and *E*), Sox9 is only specifically expressed in Epi-3 (Fig. 2*C* and Dataset S5). It is much lower in the other subtypes, including Epi-8, and is low in cells named Sox9_prolif in Garcia-Alonso et al. We therefore renamed this population as OLFM4_prolif. Immunostaining showed the presence of cells that are EPCAM+ and MKI67+ (*SI Appendix*, Fig. S4*H*). Pseudotime and velocity analysis (see below Fig. 3*C* and *SI Appendix*, Fig. S7*B*) suggests that Epi-7 and 8 are progenitor cells for the epithelial compartment.

Epi-9 is named tissue-resident memory T cell and highly expresses RUNX3 and CD69, two tissue-resident T cell markers (Fig. 2*C* and Datasets S6 and S7). It resembled a tissue-resident memory T cell population that we previously identified in the fallopian tube (*SI Appendix*, Fig. S4*G*) (38). We confirmed the presence of $EPCAM^+$ and $RUNX3^+$ cells in the uterus (*SI Appendix*, Fig. S4*H*).

In summary, our combined analysis recovered five of the previously identified epithelial cell subtypes, and renamed Sox9_Prolif to OLFM4_Prolif. Further, we identify three additional subtypes: a myoepithelial cell (Epi-7), a tissue-resident T cell (Epi-9), and a $MUC16^+$_Glandular_secretory population (Epi-2).

Identification of a Lymphatic and Eight Blood Endothelial Cell Subtypes. The cyclical shedding and regeneration of endometrium is essential for female reproduction in primates, and relies on angiogenesis, the process of new blood vessel formation (3, 39). Our initial analysis of five samples identified two main endothelial cell types: LECs and blood endothelial cells (Fig. 1 *B*–*D*), which were further corroborated in our 20-sample joint analysis (Fig. 2*A* and *SI Appendix*, Fig. S5). Given the limited number of LECs, we focused our reclustering efforts on the ~16.6K blood endothelial cells from all 20 samples, identifying eight blood endothelial cell subtypes: Endo-1 through Endo-8. These subtypes include arterial, three postcapillary venule populations (PCVs, transitioning PCVs, and activated PCVs), high endothelial venules (HEV), Tip, Capillary, and EndoMT cells, which are endothelial cells undergoing mesenchymal transition (Fig. 2 *B* and *C*, *SI Appendix*, Fig. S5*C*, and *Summaries and Discussions* for cell stability and annotation details). While most of the eight subtypes correspond to those described in Tan et al. EndoMT is new. They are characterized by high expression of mesenchymal markers such as CD44, ACTA2, ZEB2, MYH11, STEAP4, and Col1A1/2 (Fig. 2*C*), and reduced expression of classic endothelial markers [PECAM1 (CD31), TIE1, and VWF] (Datasets S6 and S7). Notably, EndoMT cells are the only endothelial cell subtype that expresses both ESR1 and PGR (*SI Appendix*, Fig. S2*C*), making it responsive to direct estrogen and progesterone signaling. However, its developmental origin and function are still unknown.

Identification of Ten Immune Cell Subtypes. Immune regulation is vital for normal uterine function, with immune cells present across all uterine compartments throughout the cycle, notably increasing in the endometrium during the late secretory phase (4, 5, 39). In this study, we identified 10 immune cell subtypes: Imm-1 to Imm-10 (Fig. 2 *B* and *C* and *SI Appendix, Summaries and Discussions*). They include innate immune cells such as Mast, Macrophages, and NK cells, as well as adaptive immune cells such as CD8 & CD4 T cells, T regulatory (TReg) cells, B cells, and innate lymphoid cells (ILCs). In addition, we identified two unexpected populations: fibrocytes (Imm-2), which may be essential for tissue repair, and tissue-resident common lymphoid progenitor (Imm-10), which could facilitate a rapid response to local immune changes in the uterus.

Potential Progenitors in Epithelium, Endothelium, and Stromal Compartments. Past studies have shown that the uterus harbors progenitor cells in different tissue compartments, including epithelial, endometrial mesenchymal stem cells, side population cells, and bone marrow stem cells (40–45). However, it has been difficult to study cell dynamics in vivo, as it requires complex experiments such as lineage tracing or imaging of cell type-specific markers. Here, we sought to understand the potential lineage relationship among uterine cell subtypes by performing RNA pseudotime analysis (46). This is followed by "velocity" analysis (47), which leverages the spliced and unspliced transcripts from RNA sequencing data to infer the cells' future states. By inferring the relationship among cell subtypes, we provide an estimated upstream-downstream ordering of some cell types, potentially reflecting their progenitor-descendant status during differentiation.

First, by projecting all five major cell types jointly and inferring their differentiation trajectories using *Monocle3* (46) (Fig. 3*B* and *SI Appendix*, Fig. S7*C*), we observed that Stro-11 (stromal progenitor) appears as a progenitor for other stromal cells. Stro-4 (tissue-resident vascular progenitor) acts as a bridge into the blood endothelial cells, while Epi-8 (OLFM4_Proliferative) acts as the progenitor of other epithelial cells. Epi-6 (Gland_sec-progenitor B) is the progenitor of Epi-1 (ciliated cells). The overall flow from the stromal to the endothelial and the epithelial cells is consistent with the global velocity analysis (*SI Appendix*, Fig. S7*A*), suggesting that Stro-11 is a global progenitor, and that human uterus stromal cells have the potential to undergo a mesenchymal-to-epithelial (MET) transition to contribute to the regeneration of epithelial and endothelial compartments during the normal cycle, an observation only seen in postpartum uterus in mice (48–50).

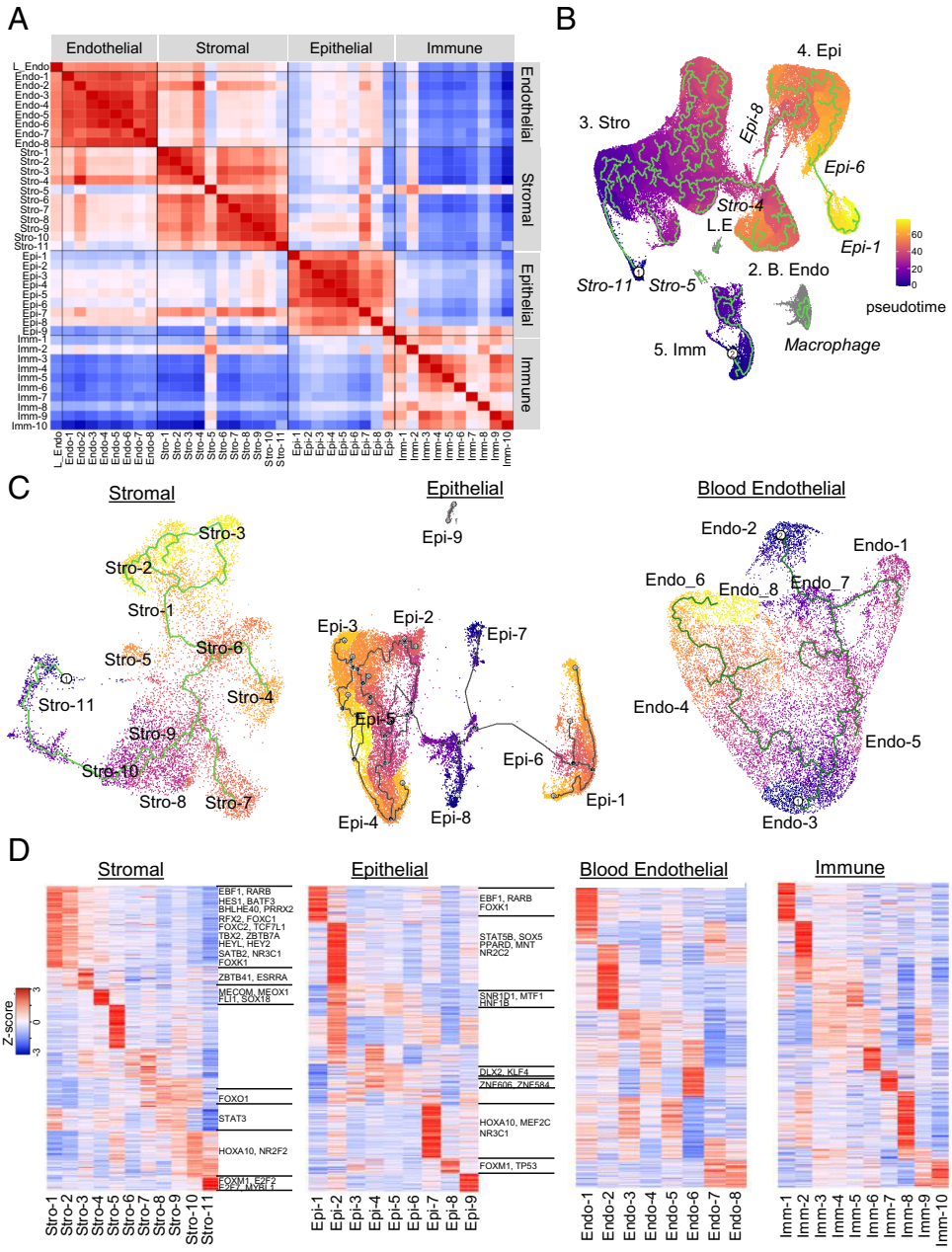


Fig. 3. Potential differentiation trajectories inferred from subtype correlations and pseudotime analysis. (*A*) Global comparison of 39 subtypes to highlight dual-character subtypes that may "bridge" across major cell types. Shown in the heatmap of 39-by-39 pairwise rank correlation values among the 39 subtype centroids. (*B*) Joint UMAP projection of cells in all five major cell types, with pseudotime from Monocle3 shown in a color gradient (blue-early; yellow-late), and inferred trajectories shown as green lines. (*C*) Pseudotime-based inference of differentiation trajectories for stromal, epithelial (including the ciliated cells), and blood endothelial cells. Note: the Pseudotime analysis for stromal cells shown here is based on subsampled cells, with UMAP the same as in Fig. 2*B*. Parallel results for all stromal cells are in *SI Appendix*, Fig. S7*A*. (*D*) SCENIC-based TF activities differ across stromal, epithelial, blood endothelial, and immune cell subtypes. Shown are TF-subtype centroid heatmaps for per-TF standardized values (data in Dataset S12). Color range is (−3.2, 3.2) for stromal cells. and (−3, 3) for the other three.

The intermediate or "transitional" subtypes identified by velocity and pseudotime analysis, including Stro-4, Epi-8, and Epi-6, were also supported by the pairwise similarity patterns among all 39 cell subtypes (Fig. 3*A*). In addition to the three identified transitional cell types, we find many other "dual-character" cells, such as Endo-2 (EndoMT, with similarities to stromal cells), Epi-7 (myoepithelial, with resemblance to epithelial and stromal cells), Epi-9 (tissue-resident memory T cell, similar to immune cells), and Imm-2 (fibrocytes, similar to stromal cells, especially Stro-5) (Fig. 3*A*). While not resolved in the global projection in Fig. 3*A*, they appeared as outlying clusters and/or source populations in the zoomed-in projections for individual cell types (see below).

Next, we focused on inferring developmental trajectories within each of the four compartments (Fig. 3*C* and *SI Appendix*, Fig. S7*B*). Among the 11 stromal subtypes (Fig. 3 *C*, *Left*), Stro-11 appears to differentiate into Stro8-9-10 (the dS subtypes), which then branch to Stro-7 (C7 Fibroblast) and Stro-6 (uSMC). Stro-6 in turn bifurcates to Stro-4 (the bridge toward the blood endothelial cells) and to Stro-1 (PV), which gives rise to Stro-2 (the other PV MYH11 population) and Stro-3 (STEAP4) as two terminal states.

Among the blood endothelial cells, Endo-3 (PCV) appears as a source population for two divergent trajectories, the first generating Endo-4 (transitioning PCV), and then Endo-6 (activated PCV), and the second to Endo-7 (Tip) by way of Endo-5 (Capillary). Tip cells further bifurcate to Endo-1 (Artery) and Endo-8 (HEV). Endo-2 (EndoMT) appears as an outlier and has endothelial-stromal dual characteristics (Fig. 3 *C*, *Right*).

Among the epithelial subtypes, Epi-8 (OLFM4_Prolif) and Epi-7 (myoepithelial) appear to be pan-epithelial progenitors (Fig. 3 *C*, *Center*). They differentiate to two directions: Epi-6 (Glandular secretory-progenitor B), which generates the ciliated cells (Epi-1), and Epi-5 (Glandular secretory-progenitor A), which flows to Epi-2, 3, and 4 as alternative end states (Fig. 3 *C*, *Center*). Epi-9 (similar to immune cells) is an isolated cluster in this projection (Fig. 3 *C*, *Center*).

In summary, pseudotime and velocity analysis suggested both compartment-specific (Endo-3, Stro-11, Epi-8, 5-6, and -7) and cross-compartment progenitors (Stro-11, Epi-8, Stro-4). These predicted progenitor populations may constitute a coordinated program, involving a hierarchy of progenitors, for optimal tissue maintenance and regeneration.

Gene Regulatory Networks (GRNs) in Specific Cell Types. After identifying multiple progenitors and their differentiation trajectories, we next focused on identifying the GRNs that govern these processes. The transcriptional state of a cell arises from complex interactions of TFs and their downstream target genes, yet a major limitation of scRNAseq is that many of the TFs are expressed at low levels. To increase the sensitivity of GRN analysis we applied SCENIC (Single-Cell rEgulatory Network Inference and Clustering) (51), which leverages the (co)expression patterns of downstream genes to estimate the overall activity of individual TF-activated "regulons". We produced per-cell TF activity scores and performed DE analysis among cell subtypes within each of the four major cell types. This led to the identification of 494, 591, 575, and 560 DE TFs in the stroma, epithelial, endothelial, and immune cells, respectively (Fig. 4*A* and Dataset S12).

The reliability of GRN network predictions is underscored by their ability to both recapitulate known biology and uncover new insights into gene regulation. For instance, the decidualization of endometrial stromal cells (52) is known to play a crucial role in embryo implantation, modulating maternal immune response, and controlling placentation (53, 54). Key TFs implicated in this process include STAT3 (55), FOXO1 (56), HOXA10, and NR2F2-HAND2 (57). Here, we find that the STAT3 activity is highest in Stro-8 and Stro-9, but low in Stro-10 (Fig. 4*A* and Dataset S12), suggesting that STAT3 signaling is not constitutively on in all decidualized cells. In contrast, FOXO1 activity is high in all 3 decidualized cell populations, while HOXA10 and NR2F2 are low in Stro-8, higher in Stro-9, and highest in Stro-10 (Fig. 4*A* and Dataset S12). Among the classic decidualization factors, we identify additional TFs that enrich in different stages of decidualization: early dS1 (PBX3; NFKB2), mid dS2 (MEIS1, MAFB, XBP1), and late dS3 (CERS4, MXD4, ING4) (Dataset S12).

Furthermore, the GRNs can be used to further distinguish closely related subtypes and contribute to their specialization. For example, within PV cells (Stro-1 to 3), certain TFs like EBF1, RARB, HES1, SATB2, NR3C1, FOXK1, and BATF3 are high across all subtypes, while others are more subtype-specific: Stro-1 shows high levels of BHLHE40, PRRX2, RFX2, FOXC1 & 2, ZBTB7A, and TCF7L1 (Fig. 4*A* and Dataset S12); Stro-2 is characterized by high TBX2, HEYL, and HEY2 levels (Fig. 4*A* and Dataset S12); and Stro-3 is distinguished by high ZBTB41 and ESRRA levels (Fig. 4*A*). Similarly, among the epithelial cells, some TFs are shared across glandular secretory cells (Epi-2, -5, and -6): NR1D1, MTF1, and HNF1B, while Epi-2 can be distinguished by high activities of STAT5B, SOX5, PPARD, MNT, and NR2C2, Epi-5 by DLX2 and KLF4, and Epi-6 by ZNF606 and ZNF584.

Finally, we turned to GRNs to identify novel regulators for uterine tissue progenitor cells. In Stro-11, a likely multipotent progenitor, we find high activities for FOXM1, E2F2, 7, 8, and MYBL1 and 2 (*SI Appendix*, Fig. S8*A*), TFs not extensively characterized in the uterus. For Epi-7 and -8, two epithelial cell progenitors, we identify HOXA10, MEF2C, NR3C1 as high in Epi-7 (*SI Appendix*, Fig. S8*B*), and FOXM1, TP53 in Epi-8 (*SI Appendix*, Fig. S8*C*). Notably the Human Protein Atlas database (58) shows HOXA10, MEF2C, and NR3C1 proteins localized at the gland base, aligning with where we mapped Epi-7 (*SI Appendix*, Fig. S8*B*). Consistent with GRN predictions, FOXM1 is found in 1 to 2 cells per epithelium gland along with TP53, while FOXM1 was rarely detected in neighboring rare stromal cells (*SI Appendix*, Fig. S8 *A* and *C*). Finally, Stro-4, a vascular progenitor stromal cell has high activities of classic stromal cell as well as endothelial cell lineage TFs: MECOM (59), MEOX1 (60, 61), and FLI1 and SOX18 (62, 63) (Fig. 3*D*), suggesting that these GRNs may be key regulators in the stromal-to-endothelial cell transition.

In summary, the SCENIC-based TF analysis enables us to identify TFs crucial for maintaining cell states or driving differentiation, and pinpoint where they are specifically expressed. Many of the knockout-based perturbation experiments in the uterus rely on broad Cre drivers, such as Progesterone Receptor-Cre Recombinase, which is expressed in multiple cell types. With cell type-specific TF activity estimates we can target the cells more precisely, to pinpoint the cell of origin of specific defects and understand their cooperation in tissue development and maintenance.

Differential Abundance of Cell Populations across Cycle Phase. Endometrium remodeling involves both changes in tissue composition (i.e., cell number) and altered cell state across the menstrual cycle (see below), which we examined by comparing across cycle phases within each of the 39 cell types (Fig. 2*D*). The per-sample, per-subtype cell counts and relative abundance are in Dataset S4*B*. For stromal cells, particularly in endometrium, Stro-1-4 peak in the early secretory (ES) phase. Curiously, decidualized stromal cells exhibit distinct peak timings: early decidual cells peak in the ES phase, aligning with progesterone priming; dS2 cells peak in the Mid/Late Secretory phase and decline in the Proliferative phase; while senescent dS3 cells accumulate in the Mid/Late Secretory phase and peak in the Proliferative phase. Finally, the stromal progenitor Stro-11 is highly enriched in the endometrium and peaks during the Proliferative phase, indicating its role in active tissue remodeling.

In the epithelial compartment, the glandular secretory populations, Epi-2, -5, -6, all peak in the secretory phases (Fig. 2*D*), while Epi-1, ciliated cells, had more variability across the cycle (Fig. 2*D*). Notably, the two epithelial progenitor cells displayed distinct peak times: Epi-8 was elevated during both the proliferative and ES phases, whereas Epi-7 was enriched specifically during the secretory phase. This suggests that these progenitor cells are involved in different processes that are regulated by the timing of their activation (Dataset S4*B*).

For blood endothelial cells, PCV (Endo-3) and tPCVs (Endo-4) are detected in both endometrium and myometrium (Fig. 2*D*), with PCV dominating the myometrium in the late secretory phase. In contrast, aPCVs (Endo-6) are the most abundant in endometrium and in the late-secretory phase, when it is the lowest in myometrium. These patterns suggest differential roles of EC-aPCVs in myometrium and endometrium. Endo-8 (HEV) cells are much more abundant in endometrium than myometrium (Fig. 2*D*) and enrich in both the proliferative and the mid-late secretary phases. This pattern is the opposite of that of Endo-6, aPCV, which is enriched in the ES phase. Finally, TIP (Endo-7) cells are mainly found in the endometrium, and most abundant in the proliferative phase (Fig. 2*D*). This is consistent with Tip cells playing a pivotal role in sprouting and creating new blood vessels in the proliferative phase.

Among the immune cells, the Mast and Fibrocytes were more prevalent in myometrium and particularly abundant in the late secretory phase (Fig. 2*D*), consistent with roles in embryo implantation (12, 64). In contrast, CD8, CD4, and Treg are maintained throughout the cycle, except they are reduced in the myometrium in the late secretary phase (Fig. 2*D*). Similarly, NK cells and Macrophages have a consistent proportion in the endometrium, with a transient increase in proliferative-to-Early-Secretory and Mid/Late secretary, respectively (Fig. 2*D*). Finally, the ILCs are enriched in the proliferative and early-mid secretory phases in endometrium, and decrease afterward (Fig. 2*D*). Collectively, these data highlight the intricate cellular dynamics across compartments that regulate uterine function throughout the menstrual cycle.

Phase-Dependent Regulation of Cell States in Individual Cell Types. Next, to examine the changes in cell states across the menstrual cycle, we performed differential expression (DE) analysis within individual cell types across cycle phases. The centroids for the three broadly defined phases—proliferative (P), ES, and mid-late secretory (MLS)—are in Dataset S13. By DE

analysis of P-vs-ES and ES-vs-MLS, we identified tens to hundreds of DE genes in most uterine cell types (*SI Appendix*, Fig. S9*A* and Dataset S14). The corresponding gene ontology enrichment results, reflecting each cell subtype's phase-dependent shifts of functional state, are in Dataset S15. In all, 1,308 and 2,190 genes are significantly phase-regulated (FDR < 0.05, Fold-change > 1.6) in at least one subtype of the stromal and epithelial cells, respectively (Fig. 4*A* and Dataset S16).

Within the stromal or epithelial compartment, the phase-regulated genes identified in one subtype are often detected in other subtypes (Fig. 4*A*), as also shown by the extensive overlap among the DE gene lists (*SI Appendix*, Fig. S9 *B* and *C* and Dataset S16). The coordinated regulation across multiple cell types reflects their shared tissue environment, with concerted changes driven by paracrine or juxtacrine signaling to ensure robust homeostasis during menstrual cycle. For instance, among genes with uniform expression across multiple stromal subtypes in the proliferative phase are those associated with cell proliferation (e.g., MDK, MAP2K2) and tissue regeneration (e.g., MFAP2, ECM1, MMP11) (see the Selected-gene tabs in Dataset S16). Likewise, some proliferative-high genes are shared across multiple dS cells, Stro-8 to -10, including those involved in immune surveillance (e.g., CXCL12, LGALS3BP, SPON2). In contrast, some other phase-regulated genes are specific for one or two subtypes. For example, among proliferative phase-high genes are those that promote angiogenesis (e.g., CCN3, CCN5, CAV1, VEGFB), and they are specific to Stro1-3 (Fig. 4*A*). Such patterns of both broadly shared and cell type-specific phase-regulation is also evident in the epithelial cells (Fig. 4*A* and Dataset S16).

While prior studies of cycle-dependent regulation mainly relied on bulk analysis, our data support cell type-specific dissection of functional changes during the cycle, including potential interactions across compartments and across phases. For instance, both the Sox9-glandular secretory cells (Epi-3) and Glandular progenitor A (Epi-5) highly express Indian Hedgehog (IHH) in the proliferative phase, while its receptor, PTCH1, is only specifically expressed in myoepithelial progenitor cells (Epi-7), decidual stromal cells (Stro-9 and -10), and stromal progenitors (Stro-11) (Fig. 4*B*). In another example, IL6 is highly specific to the luminal epithelial cells (Epi-4) and mainly in the ES phase, yet its receptor, IL6R, broadly express in all epithelial cells and constitutively in all phases (Fig. 4*B*). A reverse example is LIF, which is broadly expressed across epithelial subtypes and moderately higher in the ES phase, yet its receptor, LIFR, is much more specific: in Stro-3, the STEAP4 PV cells and Epi-8, the OLFM4_proliferative cells, and only during the ES phase. As such, these phase- and cell type-specific patterns can be leveraged to dissect the dynamic interplay among uterine cells over the menstrual cycle and inform the targeting strategies of perturbation experiments.

WNT and Insulin-LikeGrowth Factor (IGF) Pathways Likely Promote Endometrial Regeneration in the Proliferative Phase. We extended the phase- and subtype-specific analyses to biological pathways that have been previously implicated in endometrial proliferation, including Wingless-related integration site (WNT), IHH, and IGF (12, 65–69). For WNT signaling, three of the WNT ligands—WNT2, WNT4, and WNT5A—are highly expressed in Stro-8 to Stro-11, albeit with different phase-dependent patterns: WNT2 and WNT5A are high at all phases, while WNT4 in Stro-8 is specifically high in the MLS phase (Fig. 4*B*). Another ligand, WNT7A, is specifically expressed in Epi-4, the luminal epithelial cells, with minimal expression in Epi-6, glandular secretory B, only during the proliferative phase (Fig. 4*B*). These compartment-specific expression patterns in humans are consistent with those reported in mice (70) (WNT2 was not examined in mice). We then examined downstream components of WNT signaling, starting from the cell surface coreceptors, Frizzled (FZD1-8, FZD10) and LRP5-6, which activate Disheveled, leading to the inhibition of destruction complex (consisting of AXIN1/2, APC, and GSK3B). Subsequently, β-catenin (CTNNB1) accumulates in the nucleus and interacts with TCF/LEF1 TFs (e.g. TCF7L2). In our data, these downstream genes show little phase variation (Fig. 4*B*), consistent with their potential regulation on a posttranscriptional level. Other than activators, WNT signaling can be inhibited by secreted Frizzled-related proteins (SFRPs) or the Dickkopf proteins (DKKs). SFRP and DKK genes are highly expressed in Stro-5 to Stro-11, as well as Epi-7, and predominantly in the proliferative phase (Fig. 4*B*), thus these cells may be WNT unresponsive due to high inhibitions. The predicted WNT inhibition in Epi-7 suggests that the main progenitor cells contributing to regenerating endometrial lining are likely Epi-8, and this is consistent with the rise in the proportion of Epi-8 cells in the proliferative phase (Fig. 2*D*). In sum, WNT ligands in stromal (WNT2, WNT4, WNT5A) and epithelial cells (WNT7A) have both constitutive and phase-specific patterns, and likely have wide-ranging effects in multiple compartments (epithelium, endothelium, and stromal) in the endometrium.

In addition to WNT, IGF signaling in mice has been implicated across all cycle phases (25, 71–73); and its dysregulation has been linked to endometrial hyperplasia and cancer (73, 74). Despite its importance in both mice (25, 72, 73) and human (75, 76) endometrial processes, the cell origins of IGF ligands and their target cell types remain poorly understood. In our data, IGF1 is highly expressed in Stro-5 to -11, and specifically in Epi-7, while IGF2 is high in Stro-8 to -11 and, at a lower level, in Epi-7 (Fig. 4*B*). Notably, IGF1 tends to be broadly expressed across all three cycle phases, while IGF2 is high in the proliferative phase (Fig. 4*B*). Their receptors, IGF1R and IGF2R, are broadly expressed over phases and cell types, with IGF1R mainly expressed higher in epithelial populations (Fig. 4*B*), and IGF2R somewhat higher in Endo 7-8 (*SI Appendix*, Fig. S10). These findings suggest that, like in mice, IGF may carry out phase-dependent roles.

NOTCH, IGF, and RA Regulate Various Physiological Processes of the Secretory Epithelium. The secretory phase of the menstrual cycle is characterized by three processes: vascular remodeling, decidualization of stromal fibroblasts, and the transformation of glandular epithelial cells into a more secretory phenotype. We examined their signaling pathways in turn.

In mice, angiogenesis and vascular remodeling involves Notch function, mediated by five ligands (JAG1, JAG2, DLL1, DLL3, DLL4) and four receptor paralogs (NOTCH1-4). JAG1 appears to be the primary Notch ligand in our data and is highly expressed in Stro-1 to -4, Epi-1, and Endo-1 and -7 (arterial and tip cells, respectively) (*SI Appendix*, Fig. S10). NOTCH1-4 are compartment-specific: NOTCH1 and NOTCH4 in endothelial cells (mostly in Endo-1 and -7), NOTCH2 in all epithelial cells, and NOTCH3 in stromal cells (mainly Stro1-4) (*SI Appendix*, Fig. S10). Across cycle phases, JAGs and NOTCHs are maintained throughout.

Stromal cell decidualization in mice is regulated by progesterone and requires crosstalk between epithelium and stromal cells. Progesterone in epithelial cells induces SOX17, which drives IHH expression (69, 77). IHH interacts with its receptors PTCH1-2, leading to the activation of NR2F2 and downstream targets BMP2, WNT4, and HAND2 (69, 78, 79). In our data for the human uterus, IHH is highly expressed during the proliferative phase in Epi-3 and Epi-5 to -9, while its receptor, PTCH1, is highly expressed in Stro-5, and Stro-8-10 during the same phase (Fig. 4*B*), suggesting a potential role for IHH in proliferative phase rather than a direct role in decidualization in the secretory phase. Unlike IHH ligand, several IHH downstream genes (FOXO1, WNT4, and HAND2) are expressed higher in Stro-8-10, decidualized stromal cells, in the secretory phase (Dataset S16). NR2F2 is uniformly expressed across all phases (Fig. 4*B*), implying that upstream signals other than progesterone and IHH may regulate NR2F2. Indeed, in cell lines and

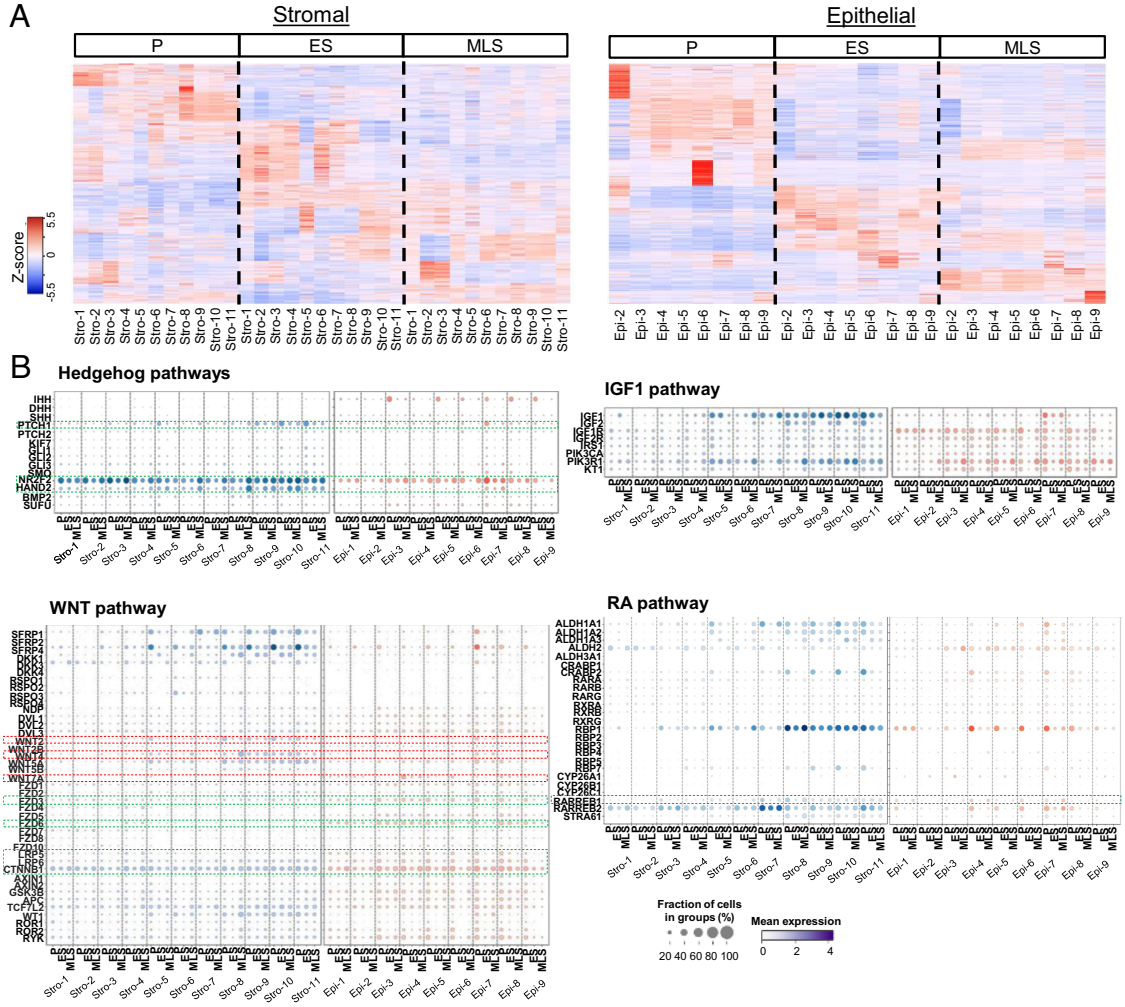


Fig. 4. Cross-phase DE analysis for individual cell subtypes and for select signaling pathways. (*A*) Phase- and subtype-resolved centroid heatmaps for stromal (*Left*) and epithelial (*Right*) cells. P: proliferative; ES: early secretory: MLS: mid-late secretory. Shown are per-gene standardized values for 1,308 and 2,190 genes that are significantly phase-regulated (*Materials and Methods*). Centroids are ordered first by phase then by subtype. Genes are ordered loosely by k-means clustering. (*B*) Cross-phase and -subtype patterns of relevant genes in four of the known signaling pathways: Hedgehog, WNT, IGF, and RA. Shown are dot plots ordered, from *Left* to *Right*, first by subtype then by phase. Complementary plots for endothelial and immune cells are in *SI Appendix*, Fig. S10 *A–D*.

model organisms NR2F2 can be induced by insulin/IGF and/or retinoic acid (80–82), suggesting that IGF and RA pathways may activate NR2F2 in the absence of IHH during the secretory phase, thereby triggering a conserved network of decidualization target genes in humans. In mice, the insulin, IGF, and RA pathways have been implicated in uterine tissue homeostasis (25, 83). For example, mice lacking either insulinreceptor (INSR) or IGF1R exhibit a partial decidual response, with a more severe defect in IGF1R knockout models (25), and mice with INS-IGF1R double knockout completely lose the decidual response (25). Additionally, RA signaling is essential for decidualization in other mammals (83). In our data, INSR and IGF1 are expressed through all phases and across most stromal cell types (Dataset S16), while IGF2 is more variable. Their receptors, IGF1R and IGF2R, along with downstream targets IRS1 and AKT serine-threonine protein kinase family, are specifically high in both proliferative and M/L secretory phases in Stro-8-10, the cells undergoing decidualization. Taken together, we speculate that downstream targets of IHH may be activated in decidualized cells by other upstream signals such as RA and IGF. Consistent with this, data in humans has shown growth hormone supplementation induces IGF-1 and improves follicular genesis, oocyte maturation, and embryo implantation (84).

Finally, epithelial cells in the secretory phase transition to a more secretory-like phenotype, and rely on a low WNT environment and Notch activation (85). The antagonistic roles of these two pathways are crucial for specification of secretory versus other cells in not only the intestine but also in the uterus, where they have been implicated in ciliated versus secretory cell specification (12, 86, 87). In our data, obligate WNT coreceptors LRP5/6 are low in the ES phase across many epithelial cell subtypes (Fig. 4*B*), likely underlying WNT-resistance in the secretory phase. Meanwhile, NOTCH is constitutively expressed in all phases. The steady expression of NOTCH and low expression of LRP5/6 in the ES phase may be sufficient to shift the balance between WNT and NOTCH, promoting a secretory-like phenotype by activating key TFs such as KLF4, which is high in the secretory phase in our data (Dataset S16).

Population Shifts in Postpartum and Postmenopausal States. While constructing this reproductive age uterus cell atlas, we had the opportunity to analyze a rare postpartum uterus sample (Donor-7, 1 wk postdelivery) and the myometrium from a postmenopausal uterus (Donor-6, age 52), allowing us to explore tissue composition changes in these distinct states.

At the level of major cell types (Fig. 4*A*), the postpartum sample showed dramatically expanded immune cells and much reduced stromal cells, consistent with the immune cell involvement in tissue repair as the uterus returns to normal after pregnancy. The reverse trend is seen in the postmenopausal sample: much expanded stromal cells, and significantly reduced immune and endothelial cells, indicating a more stable tissue state without the cyclic turnover or rapid regeneration. Both samples had few epithelial cells. Furthermore, the lymphatic endothelial population is significantly expanded in the postpartum sample, as would be expected for tissue repair.

We then evaluated the relative proportions of cell subtype within each compartment (Fig. 4*B*). In the postmenopausal sample. Stro-4 (vascular progenitor), Endo-2 (Endo MT), and Endo-6 (aPCVs) are much more abundant, indicating subtype-specific postmenopausal alterations in vasculature. The Stro-1:Stro-2 ratio (i.e.,

between the two types of MYH11 subtypes, for ECM secretion and angiogenesis, respectively) is increased, possibly contributing to uterine tissue fibrosis observed in the aging uterus. Finally, it has been shown that during uterine aging, the ability of endometrial stromal cells to decidualize decreases gradually (88). And here we find that the early dS cells (Stro-8) are significantly reduced, while later dS cells, Stro-10, are lost in the postmenopausal sample.

During the postpartum period, the uterus undergoes involution to return to prepregnancy size. This process involves apoptosis, proliferation, and autophagy of terminally differentiated cells, which are then regenerated (5). In our data, the postpartum uterus exhibits not only a lower proportion of stromal cells (Fig. 4*A*), but specifically much lower abundance of Stro-6, the uSMCs. Interestingly, Stro-5 (RUNX3+ fibroblast) is expanded in the postpartum uterus, Since RUNX3 expression in fibroblasts inhibits proliferation, the expansion of Stro-5 may act to prevent the overproliferation of other stromal subtypes following pregnancy.

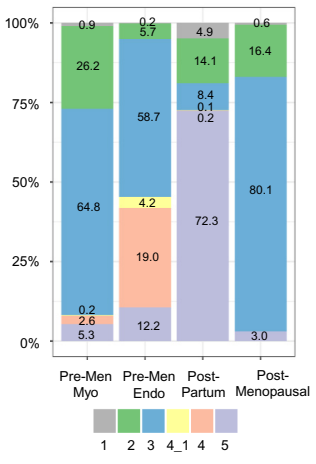
In both the postpartum and postmenopausal uterus, Stro-7 (C7 fibroblasts) is increased. Other notable changes are the reduction of Imm-2 (fibrocytes) in the postpartum sample, and the reduction of Imm-10 (hematopoietic progenitors) in the postmenopausal sample. These patterns need to be replicated in additional samples, and their functional impacts require further validation.

Discussion

Investigating the cellular composition and dynamics of the human uterus has been challenging, partly due to cyclic changes in the functional states of many cells, their interactions, and the turnover of cell communities. To profile only the endometrium or the myometrium or having limited samples that fail to cover the menstrual cycle more completely, has hindered a deeper understanding of uterus physiology. In this study, we collected full thickness uterus scRNA-seq data from five donors and integrated with previously published data from 15 donors, along with centroid data from a third study. Since each of them reported a uterus cell nomenclature, we sought to examine the concordance and inconsistencies among them, and to uncover potential progenitor populations essential for uterus tissue maintenance and regeneration. For instance, among the epithelial cells, we identified two potential progenitors: OLFM4+ cell (Epi-8) and a myoepithelial cell (Epi-7). The latter bears resemblance to similar populations (*SI Appendix*, Fig. S4*G*) found in the fallopian tube epithelium (38), underscoring a shared progenitor composition across reproductive tract tissues. Similarly, among the stromal cells, we identified a tissue-resident vascular progenitor (Stro-4), a stromal cell progenitor (Stro-11), and a RUNX3+ fibroblasts (Stro-5). Furthermore, with the larger number of cells from myometrium we were able to classify PV MYH11 into two subtypes (Stro-1 and Stro-2), which appear to serve distinct roles in vascular genesis (FOXC1^{lo} CTGF^{hi}) and ECM maintenance (FOXC1^{hi} CTGF^{lo}). Among the immune cells we identify a tissue-resident common lymphoid progenitor population (Imm-10) and a fibrocyte population (Imm-2), which has immune-stromal dual characteristics. Lastly, among the endothelial cells, we identified EndoMT, which are cells undergoing EMT, and three populations of postcapillary venules: basal PCVs, transitioning PCVs, and activated PCVs. Collectively, these results led to an expanded consensus cell atlas for the human uterus.

Homeostatic mechanisms in the uterus vary across species and seem to depend on the extent of tissue repair needed (2, 4, 89). For example, in mice, the regeneration of the uterine lining during the normal cycle is maintained by compartment-specific progenitors (48, 90). However, genetic lineage tracing in mice has shown that, after parturition, some stromal cells undergo a MET transition to generate epithelial cells (48, 90), and they have recently been identified as a PDGFRA+ fibroblast population (49). Therefore, tissue homeostasis in mouse uterus differs between normal cycles and the extensive regeneration after parturition. In humans, in vivo lineage tracing is not feasible, and characterization of stem/progenitor cells has relied on in vitro assays, which has led to the identification of multiple potential stem cell pools, including an endometrial mesenchymal stem cell, a side population cell, and a bone marrow–derived stem cell with stem/progenitor-like properties (2, 9, 40–45, 91). The molecular identities of these cells, how they compare across studies, and their contributions to uterine homeostasis and regeneration remain difficult to define. To characterize the molecular identity of stem/progenitor like cells, we performed both velocity and pseudotime analysis, which led to the identification of multiple potential compartment-specific progenitors as well as a multipotent progenitor. For example, Stro-11 is likely a multipotent mesenchymal progenitor for stromal cells, epithelial and endothelial cells (Fig. 3*B*). While the compartment-specific progenitors include: Epi-7 & -8 (Myoepithelial and OLFM4-proliferative) for epithelial cells, Endo-3 (PCV) for blood endothelial cells, and Imm-10 (hematopoietic progenitors) for immune cells. Finally, we uncover evidence of within-compartment differentiation hierarchy, e.g., Epi-6 as a likely precursor of the ciliated cells, and Epi-5 (Glandular secretory-progenitors A) as upstream progenitor for Epi-2 (Glandular secretory). Taken together, we expect that the cellular atlas described here, including both endometrial and myometrial cell subtypes and their mRNA markers, represents an enhanced resource for detailed explorations into the uterine stem/progenitor cells.

Finally, our DE and pathway analyses across the cycle revealed many conserved and novel regulators of uterine tissue cell states (Fig. 4), underscoring the importance of unraveling species-specific regulation of tissue homeostasis and physiology. Furthermore, we used



B

	Premen-Myo	Premen-Endo	Postpartum	Postmeno
Imm-1	18.7	3.9	8.4	3.3
Imm-2	8.0	2.3	0.1	11.8
Imm-3	15.3	13.4	13.2	20.9
Imm-4	7.7	9.5	10.0	7.2
Imm-5	0.2	1.2	2.1	0.0
Imm-6	0.7	5.0	0.9	0.0
Imm-7	1.5	0.9	1.0	0.0
Imm-8	38.1	18.8	49.0	52.9
Imm-9	8.9	41.1	14.4	3.9
Imm-10	0.9	4.0	0.9	0.0

	Premen-Myo	Premen-Endo	Postpartum	Postmeno
Stro-1	46.6	2.2	10.0	24.4
Stro-2	18.1	4.3	9.3	1.3
Stro-3	4.3	5.1	11.6	2.6
Stro-4	2.2	1.7	0.2	10.1
Stro-5	0.2	0.3	6.5	0.2
Stro-6	9.0	15.5	2.5	31.0
Stro-7	2.8	1.1	11.3	29.1
Stro-8	0.0	5.5	2.1	0.2
Stro-9	14.3	35.9	6.7	1.1
Stro-10	2.3	26.0	39.6	0.0
Stro-11	0.1	2.3	0.2	0.0

	Premen-Myo	Premen-Endo	Postpartum	Postmeno
Endo-1	6.6	9.2	8.9	3.7
Endo-2	4.1	14.5	0.4	33.1
Endo-3	30.3	15.4	56.5	5.0
Endo-4	30.0	14.9	11.2	2.3
Endo-5	16.0	17.4	2.2	7.2
Endo-6	12.2	11.0	1.2	48.6
Endo-7	0.2	5.5	5.5	0.0
Endo-8	0.5	12.0	14.0	0.0

Fig. 5. Altered tissue composition in a postpartum and a postmenopausal uterus sample. (*A*) Relative fractions of the major cell types across four types of uterus samples: endometrium and myometrium samples from the 20 healthy donors, a postpartum sample, and a postmenopausal sample. (*B*) Relative fractions of subtypes, for immune, stromal, and blood endothelial subtypes. Data for individual samples from the premenopausal donors are in Dataset S4*B*.

this normal cell atlas to explore tissue-level changes in a postpartum sample and a postmenopausal sample (Fig. 5). The differences in cell proportions provided an early glimpse of altered tissue function in these atypical states and need to be replicated in more samples.

Taken together, the cell atlas described here provides more than just consensus cell nomenclature and markers, but also serves as a foundation to build hypothesis regarding hierarchies of progenitor cells, differentiation trajectories, and transcriptional changes across the cycle. Such knowledge can help refining in vitro differentiation protocols and enhance the understanding of interactions and dynamics of cell communities in both healthy and diseased tissue. This is directly relevant for addressing clinical challenges such as uterine factor infertility, embryo implantation failure, and recurrent pregnancy loss. For example, it has been difficult to develop a molecular signature for endometrial biopsies to predict the time when the endometrium is most receptive to embryo implantation, partly due to variations in cell composition across individuals and pathological states. Once identified, these signatures can improve IVF outcome for patients with implantation failure and RPL (92).

Materials and Methods

Ethical Approval Process for Surgical and Cadaveric Samples. Deidentified surgical tissue samples were obtained from the Reproductive Subject Registry and Sample Biorepository (RSRSR) of the University of Michigan. The protocols and procedures of RSRSR were approved by the Institutional Review Board of the University of Michigan Medical School, registered under IDHUM00125627. All subjects undergo informed consent prior to enrollment in the biorepository. Utilization of deidentified tissue samples for this study, was approved under (HUM00167998). The deceased donor samples were procured through the International Institute for the Advancement of Medicine. The use of cadaveric tissue in this research is categorized as "not regulated," per 45 CFR 46.102 and the "Common Rule," as it does not involve human subjects, and complies with the University of Michigan's Institutional Review Board requirements as such (our exempt protocol number is HUM00250736).

Human Uterus Sample Preparation. Surgical samples were collected in Hank's Balanced Salt Solution in the operating room, transported at room temperature, and immediately processed for dissociation (*SI Appendix*). Cadaveric samples were processed within 24 to 36 h postremoval (*SI Appendix*). The cycle stages for samples were determined by an OBGYN pathologist and/or hormonal data. Briefly, uterine segments were dissected into endometrium and myometrium and dissociated separately to generate single-cell suspensions (*SI Appendix*). 10X samples were processed at the core using standard protocol. See *SI Appendix, Supporting Materials and Methods* for additional details.

Data, Materials, and Software Availability. All data is deposited in GEO (GSE260658) (93). All processed data and centroids are provided in supporting information.

ACKNOWLEDGMENTS. We thank members of Hammoud and Li Labs for scientific discussions and manuscript comments. This research was supported by NIH 1DP2HD091949-01 (S.S.H.), T32-GM70449 (D.F.H.), T32-HD079342-10 (D.B.), HL144481 (B.B.M.), Open Philanthropy grant 2019-199327 (5384) (S.S.H.) and Chan Zuckerberg Foundation Grant CZF2019-002428 (A.S., E.E.M., J.Z.L., and S.S.H.).

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